UNIVERSITY OF AGRICULTURAL SCIENCES, BENGALURU

EAI-421 | Application of AI & ML in Agriculture (0+10)

Study on Critical Stages of Crop-Weed Competition

on Growth and Yield of Soybean (Glycine max L.)

TEAM:

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Presentation Overview

01

Introduction & Significance

Why weed competition matters in soybean

02

Objectives

Three targeted research aims

03

Materials & Methods

Experimental setup at ZARS, GKVK

04

Results — Growth Parameters

Plant height & leaf count analysis

05

Results — Yield Parameter

Pods per plant & ANOVA findings

06

Discussion & Conclusions

Critical period and key recommendations

Chapter I: Introduction

1. Soybean (Glycine max L.) contains ~20% oil & ~40% protein — a critical kharif crop cultivated across MP, Maharashtra, Rajasthan & Karnataka.

2. Weeds compete for light, water, nutrients and space during early vegetative stages, causing 26–84% yield reduction depending on weed density and competition duration.

3. The critical period of crop-weed competition generally spans 15–45 DAS in Indian conditions.

4. Season-long weed control is costly and labour-intensive; identifying the exact critical window optimizes resource allocation.

KEY FACTS

26–84%
Potential yield loss from weed competition

15–45
Days after sowing — critical competition window

JS-335
Soybean cultivar used in this study

RCBD
Statistical design with 5 treatments × 4 reps

Objectives of the Study

01

Evaluate Growth & Yield Impact

Study the effect of critical stages of crop-weed competition on the growth and yield of soybean (Glycine max L.).

02

Assess Key Growth Parameters

Measure the impact of weed competition on plant height, number of leaves, and number of pods per plant.

03

Identify the Critical Period

Determine the critical period of weed interference in soybean under the agro-climatic conditions of Bengaluru, Karnataka.

Materials & Methods

📍 Experimental Location

ZARS, UAS, GKVK, Bengaluru – 560 065

Lat: 13.0713°N | Long: 77.5905°E | Alt: 924 m

Red Sandy Loam | pH: 6.56 | OC: 0.49%

Available N: 445.6 | P: 46.36 | K: 243.9 kg/ha

🔬 Experimental Design

Design: RCBD — 5 treatments × 4 replications

Crop: Soybean (Glycine max L.) cv. JS-335

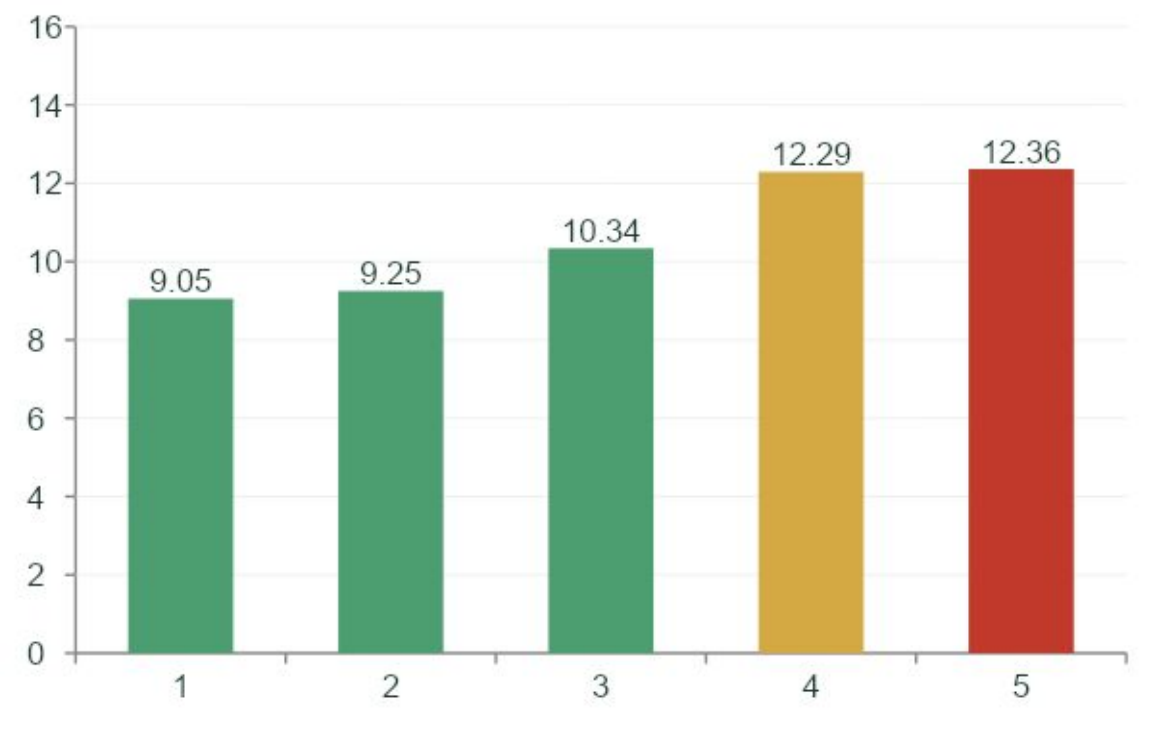
Spacing: 30 cm × 10 cm | Seed rate: 70–75 kg/ha

Period: Sown: 21/09/2024 | Harvested: 11/12/2024

Treatment Details

T1	Weed-free up to 15 DAS	
T2	Weed-free up to 30 DAS	
T3	Weed-free up to 45 DAS	
T4	**Weed-free throughout (full season)**	★ BEST PERFORMER
T5	Weedy check — No weed control	

Result 1: Plant Height at 30 DAS



ANOVA RESULT

F-calculated

3.257

F-tabulated (5%)

3.26

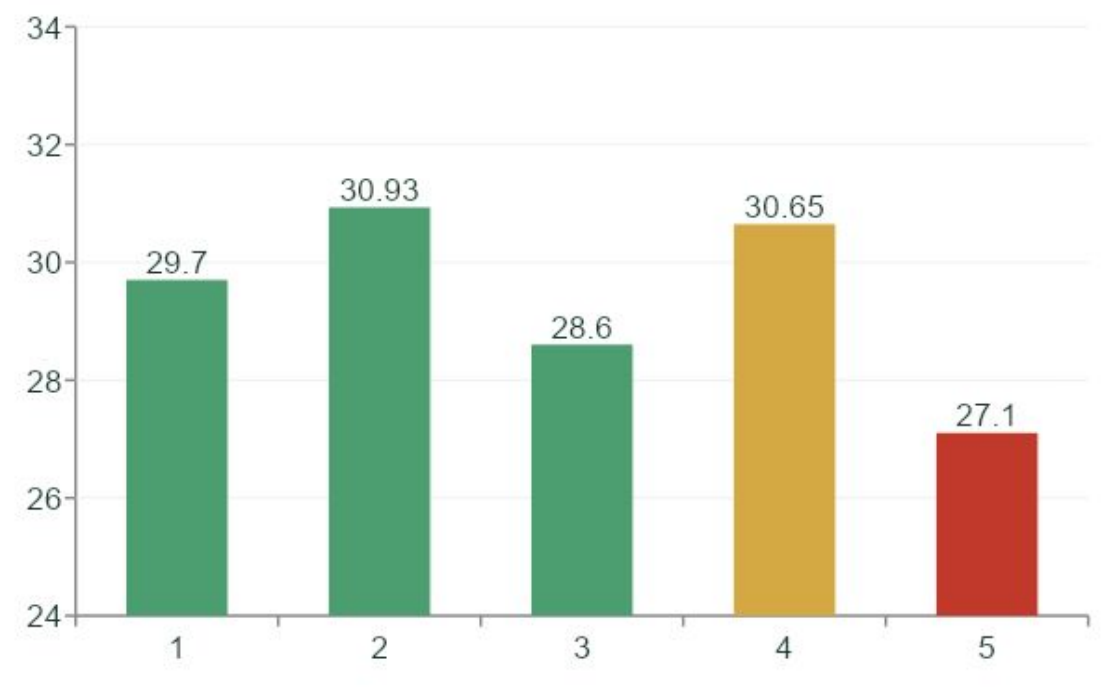
Verdict

SIGNIFICANT ✓

T4 recorded highest mean height (12.29 cm).
T1 showed lowest (9.05 cm).

F-cal (3.257) > F-tab (3.26) — Significant treatment effect exists at 30 DAS

Result 2: Plant Height at 60 DAS



KEY FINDING

NOT SIGNIFICANT

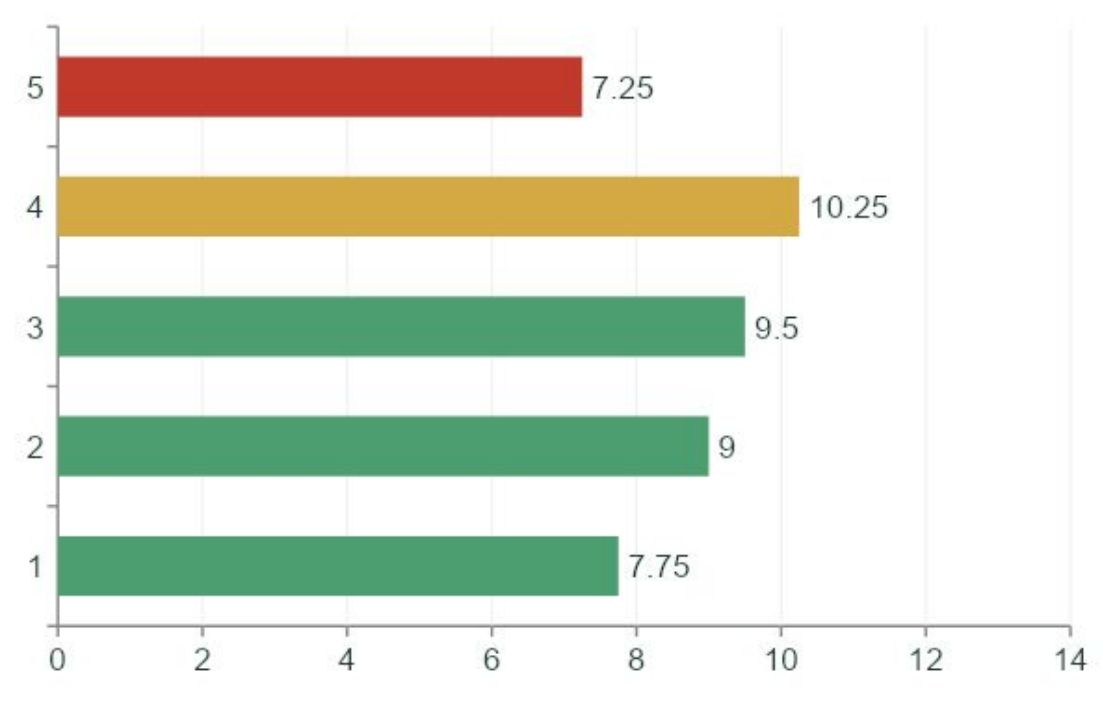
F-cal = 0.188 < F-tab = 3.26

What does this mean?

By 60 DAS, the soybean canopy closes and naturally suppresses weeds — differences in plant height across treatments converge. Early weed removal advantage is maintained in yield parameters, not vegetative height.

Vegetative height equalizes by 60 DAS — early weed management advantage shifts to yield components

Result 3: Number of Leaves per Plant



SIGNIFICANCE

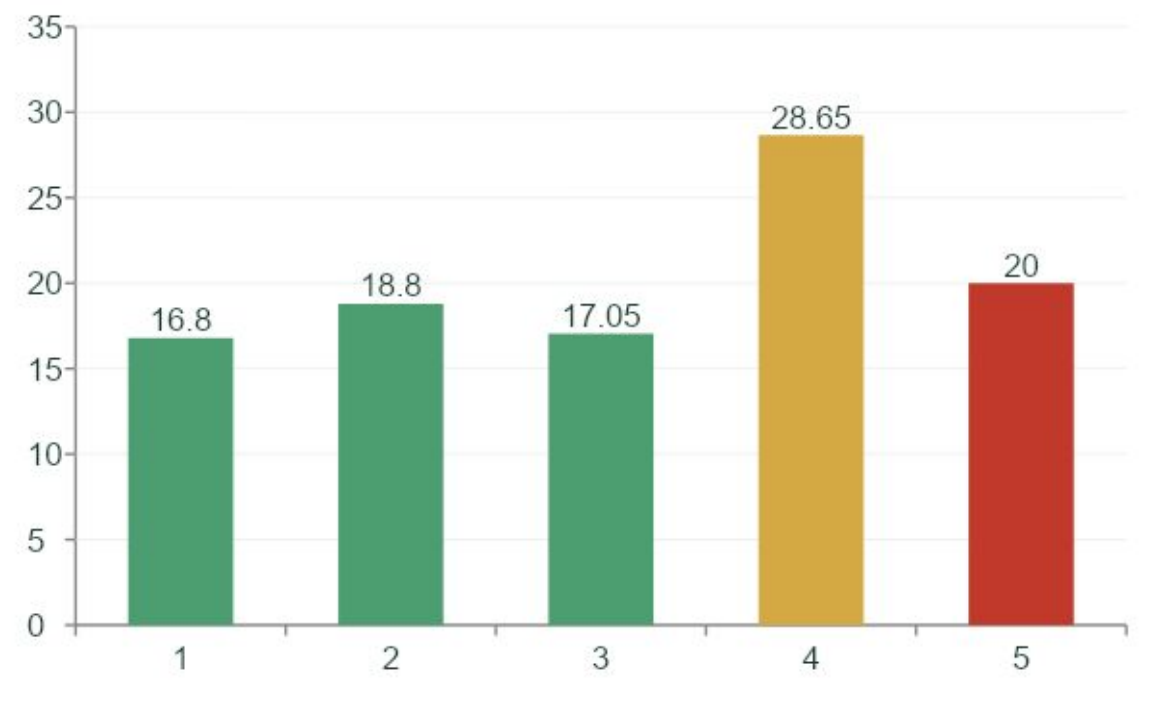
F-cal = 3.26
= F-tab (5%)

SIGNIFICANT ✓

- T4 (10.25 leaves) — highest photosynthetic capacity
- T5 (7.25 leaves) — suppressed by persistent weeds
- Leaves = source of photosynthate for pod filling

Significant differences in leaf count confirm weed pressure suppresses soybean's photosynthetic capacity

Result 4: Number of Pods per Plant ⭐ Most Significant



ANOVA RESULT

F-cal = 8.424

F-tab = 3.26 (5%)

HIGHLY SIGNIFICANT **

28.65 pods — T4 (highest)

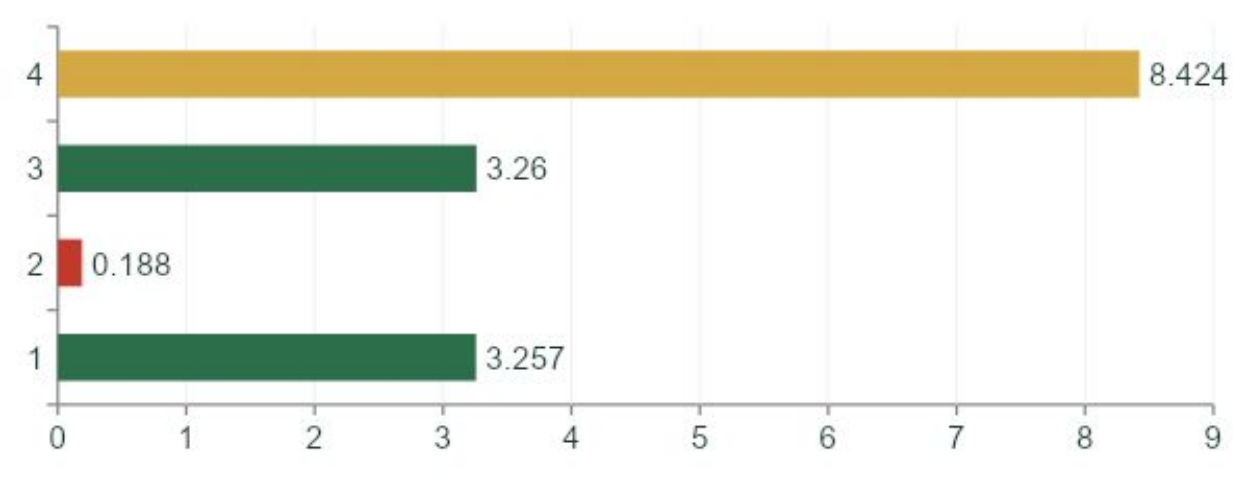
16.80 pods — T1 (lowest)

+70% gain with full-season control

Pods per plant shows the highest F-value (8.424) — season-long weed control is essential for maximizing yield

Statistical Analysis Summary — ANOVA (RCBD)

Parameter	F-Calculated	F-Tabulated (5%)	Result
Plant Height — 30 DAS	3.257	3.26	Significant ✓
Plant Height — 60 DAS	0.188	3.26	Non-Significant ✗
No. of Leaves / Plant	3.260	3.26	Significant ✓
No. of Pods / Plant	8.424	3.26	Highly Significant ✓✓



F-tab Threshold

3.26

at 5% significance level for RCBD

Bars exceeding this line are significant

Treatment Performance Summary

	T1	T2	T3	T4	T5
	Weed-free 0–15 DAS	*Weed-free 0–30 DAS*	*Weed-free 0–45 DAS*	*Full Season Weed-free*	*Weedy Check*
Ht. 30 DAS	9.05 cm	9.25 cm	10.34 cm	12.29 cm	12.36 cm
Ht. 60 DAS	29.7 cm	30.93 cm	28.6 cm	30.65 cm	27.1 cm
Leaves	7.75	9	9.5	10.25	7.25
Pods/Plant	16.8	18.8	17.05	28.65	20

★ *T4 (Season-long weed control) is the undisputed best performer across all parameters — especially pod count (+70% vs T1)*

Conclusions & Recommendations

Critical Period Confirmed

The critical period of crop-weed competition in soybean at GKVK lies between 15–45 DAS, consistent with published literature.

Season-Long Control Wins

T4 (weed-free throughout) recorded the highest pod count (28.65/plant) — ~70% more than T1 — proving the value of full-season management.

Early Control Is Insufficient

Weed removal only up to 15 DAS (T1) is inadequate; control must extend to at least 30–45 DAS to protect yield.

Vegetative Recovery

Plant height at 60 DAS shows no significant differences — the crop recovers vegetative growth, but yield (pods) suffers without continuous weed management.

Recommendation

Integrate pre-emergence + post-emergence herbicides with timely inter-cultivation within the 15–45 DAS window for cost-effective, labour-efficient weed control.

Future Scope

Multi-location trials, economic analysis, herbicide evaluations, drone-based precision weed mapping, and climate-adaptive weed management studies are recommended.

Thank You

Study on Critical Stages of Crop-Weed Competition

EAI-421 Capstone Project | BSc (Hons) Agriculture

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